

Roll No. - 123

(Please write your Enrolment No. immediately)

Enrolment No. 193 20803121

B.Tech. End Term Examination

Third Semester, January, 2023

Paper Code: HS-203

Subject: Indian Knowledge System

Time: 3 Hours

Max Marks: 75

Note: Attempt Q.No. 1 which is compulsory. In addition to Q.No. 1 attempt any one question from each unit. All questions carry equal (15) marks

Q.1 Answer any 5 of the following: (5X3 marks)

- How many types of Vedas are identified till now? State the significance of Vedas in modern times.
- What are the Vedangas? Address the role of all six Vedangas.
- Briefly explain Panini's work on Sanskrit grammar.
- How is the issue of phonetics addressed in Sanskrit language?
- What are the elements of Pancanga ?
- Indian civilization has a rich heritage in science and technology (S&T). Discuss the evidence.
- Comment on the statement, "Susruta is the father of surgery".

UNIT-I

- Comment on Yoga as a school of philosophy founded by Maharshi Patanjali. (7 marks)
- Explain the importance of eight steps of Ashtang yog. (8 marks)

Q.3 Write notes on: (3x5 marks)

- Haldi and Neem patent battle with USA
- Upanishads
- Vedic life - distinctive features

UNIT-II

- Comment on the statement, "The Sanskrit language has potential for use in NLP and AI applications". (7.5 marks)

b) What are the unique aspects of the Sanskrit grammar propounded by Panini? (7.5 marks)

- What are the key features of the Indian Numeral System and what role did it play in advancement of Science and Technology? (7.5 marks)

b) Describe the existence of robust measurement system found in ancient Indian literature in terms of quantifying length, time and weight. (7.5 marks)

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UNIT-III

- Q.6 a) Explain in detail Jantar Mantar of Raja Jai Singh Sawai. (8 marks)
- b) What are the key contributions of Aryabhatiya in the field of astronomy? (7 marks)
- Q.7 a) Explain Celestial Coordinate System, (8 marks)
- b) Write Short notes on any two of the following ; (7 marks)
- (i) Rasi
 - (ii) Naksatra and Tithi
 - (iii) Uttarayana

UNIT-IV

- Q.8 a) What do you understand by the Lost-wax casting process of making an idol? (5 marks)
- b) How can we infer ancient Indian's knowledge and use of iron and copper? (5 marks)
- c) What are the different varieties of iron listed by ancient Indians? (5 marks)
- Q.9 a) (i) Dola-yantra and (ii) Patana-yantra were used by alchemists in India. Write a brief account of these two apparatuses. (5 marks)
- (b) Wootz steel made in India thousand years back is being called the advanced material of the ancient world. Explain why and how we lost over to Europe in this aspect of business of this steel. (5 marks)
- (c) RRS (Ras-ratna-samuccaya) mentions a process of extraction of copper for Ayurvedic purposes. Explain it. (5 marks)